

Recall — percentages

A percentage is a fraction with denominator 100 — and everything else follows from that.

LESSON 55

1 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 10 ENTRY

LESSON CLOCK

10'

12'

12'

6'

What a percentage is	10 min
The three forms	12 min
Finding a percentage of an amount	12 min
One quantity as a percentage of another	6 min

LEARNING OBJECTIVES

- Convert between percentages, fractions and decimals.
- Find a percentage of a quantity.
- Express one quantity as a percentage of another.
- Recognise the percentages worth knowing by heart.

TEXTBOOK

National	pp. 154–157
Cambridge	Stage 7, pp. 96–100
Workbook	Exercise 10.1

01

Explanation

What a percentage is

$$p\% = \frac{p}{100}$$

“Per cent” means “out of a hundred”, so 37% is $\frac{37}{100}$ and nothing more mysterious.

STATEMENT	MEANS
50%	half
25%	a quarter
10%	a tenth
100%	all of it
150%	one and a half times it

PERCENTAGES EXIST TO MAKE COMPARISON EASY

Is $\frac{17}{25}$ better than $\frac{27}{40}$? As 68% and 67.5% the answer is immediate. A common denominator of 100 is the whole point.

The three forms

PERCENTAGE	FRACTION	DECIMAL
10%	$\frac{1}{10}$	0.1
20%	$\frac{1}{5}$	0.2
25%	$\frac{1}{4}$	0.25
50%	$\frac{1}{2}$	0.5
75%	$\frac{3}{4}$	0.75
12.5%	$\frac{1}{8}$	0.125
$33\frac{1}{3}\%$	$\frac{1}{3}$	0.333...

PERCENTAGE TO DECIMAL: DIVIDE BY 100

And decimal to percentage: multiply by 100. The digits move two places, and which way they move is settled by asking whether the answer should be bigger or smaller.

Finding a percentage of an amount

TASK	METHOD	ANSWER
10%	divide by 10	24
20%	double the 10%	48
5%	half the 10%	12
25%	divide by 4	60
15%	10% + 5%	36
37%	$0.37 \cdot 240$	88.8

Every row is a percentage of 240.

BUILD AWKWARD PERCENTAGES FROM EASY ONES

10%, 5% and 1% are all one division away, and 35% is $10\% + 10\% + 10\% + 5\%$. Mental percentage work is nearly all addition.

One quantity as a percentage of another

$$\text{percentage} = \frac{\text{part}}{\text{whole}} \times 100\%$$

PART	WHOLE	WORKING	PERCENTAGE
17	25	$\frac{17}{25} \cdot 100$	68%
27	40	$\frac{27}{40} \cdot 100$	67.5%
18	24	$\frac{18}{24} \cdot 100$	75%
9	60	$\frac{9}{60} \cdot 100$	15%

THE WHOLE GOES UNDERNEATH

“What percentage of 40 is 27” puts 40 in the denominator. Swapping them gives 148%, which the sense check rejects at once.

02 Worked examples**EXAMPLE 1** Find 15% of 240.

① 10% is 24.

② 5% is half of that, 12.

③ $24 + 12 = 36$

ANSWER

36

EXAMPLE 2 What percentage of 40 is 27?

① $\frac{27}{40}$

The whole underneath.

② $\cdot 100 = 67.5$

③ 67.5%

ANSWER

67.5%

EXAMPLE 3 Which is better, 17 out of 25 or 27 out of 40?

① $\frac{17}{25} = 68\%$

② $\frac{27}{40} = 67.5\%$

- ③ The first, by half a percentage point.
Impossible to see without converting.

ANSWER

17 out of 25

03 Interactive model

Put two test scores out of different totals on the board and ask which is better; the class reaches for percentages without being told to.

Percentage, fraction, decimal

25% as a fraction:

$$\frac{1}{4}$$

$$\frac{1}{25}$$

$$\frac{25}{1}$$

$$\frac{1}{5}$$

0.2 as a percentage:

2%

20%

0.2%

200%

10% of 240:

2.4

24

240

12

5% of 240:

6

12

24

48

15% of 240:

30

36

40

48

27 as a percentage of 40:

67.5%

148%

54%

75%

18 as a percentage of 24:

60%

75%

80%

133%

150% of a number is:

half of it

one and a half times it

impossible

the same

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Percentage	Foiz	Процент
Per cent	Yuzdan	Со ста
Fraction	Kasr	Дробь
Decimal	O'nli kasr	Десятичная дробь
To convert	O'girmoq	Преобразовать
Of a quantity	Miqdordan	От величины
Whole	Butun	Целое
Equivalent	Teng kuchli	Равный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

37% means:

$$\frac{37}{10}$$

$$\frac{37}{100}$$

$$\frac{100}{37}$$

37

0.75 as a percentage:

7.5%

75%

0.75%

750%

1 8 as a percentage:

8%

12.5%

18%

1.8%

25% of 240:

48

60

80

96

To write a part as a percentage you divide by:

the part

the whole

100

the difference

9 as a percentage of 60:

9%

15%

54%

6.7%

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 25% as a fraction

ANSWER $\frac{1}{4}$

2. 20% as a decimal

ANSWER 0.2

3. 0.75 as a percentage

ANSWER 75%

4. $\frac{1}{8}$ as a percentage

ANSWER 12.5%

5. 10% of 240

ANSWER 24

6. 25% of 240

ANSWER 60

7. 50% of 240

ANSWER 120

Medium · the standard question

7 PROBLEMS

1. 5% of 240

ANSWER 12

2. 15% of 240

ANSWER 36

3. 37% of 240

ANSWER 88.8

4. 17 as a percentage of 25

ANSWER 68%

5. 27 as a percentage of 40

ANSWER 67.5%

6. 18 as a percentage of 24

ANSWER 75%

7. 9 as a percentage of 60

ANSWER 15%

Hard · stretch and reason

7 PROBLEMS

1. 35% of 240 by building from 10%

ANSWER 84

2. 12.5% of 320

ANSWER 40

3. Which is better, $17/25$ or $27/40$?

ANSWER $17/25$, at 68%

4. 150% of 60

ANSWER 90

5. 1% of 240, and hence 3%

ANSWER 2.4 and 7.2

6. A test mark of 45 out of 60 as a percentage

ANSWER 75%

7. Why do percentages make comparison easy?

ANSWER Every fraction is put over the same denominator, 100

07 Homework

Homework — 5 tasks

Build awkward percentages from 10%, 5% and 1%.

1. Convert 45%, 8% and 130% to decimals and to fractions in lowest terms.
2. Find 10%, 5% and 15% of 360.
3. Find 35% of 480.
4. Write 21 as a percentage of 35.
5. Which is the better mark: 19 out of 25 or 30 out of 40?